CSE 318: Artificial Intelligence Sessional

Online 01

C2

Twin-Blank N-Puzzle

Problem Statement

Modify your existing N-Puzzle solver so that the board contains two blank cells instead of one.

For a 4×4 board, the puzzle will contain:

- tiles from `1` to `14`,
- two blank cells, both represented by `0`.

In one move, any numbered tile adjacent to either blank cell may slide into that blank cell.

The two blank cells are indistinguishable. That means swapping the identity of the two blanks should not create a different state.

Goal State

For this task, all input boards will be 4×4. For a 4×4 board, the goal state is:

```
1  2  3  4
5  6  7  8
9 10 11 12
13 14  0  0
```

Required Modification

Update your program to handle two blank cells.

During neighbor generation, consider both blank cells.

For each blank cell, check its valid neighboring cells in the four directions:

```
U, D, L, R
```

Heuristic Function

Use Manhattan distance over the numbered tiles only.

The two blank cells should not be included in the heuristic calculation.

$$h(n) = \sum_{\text{numbered tiles } t} (|r_t - r_t^*| + |c_t - c_t^*|)$$

Here, (r_t, c_t) is the current position of tile t, and (r_t^*, c_t^*) is the goal position of tile t.

Solvability

For this task, all given test cases will be solvable.

You do not need to use the old inversion-parity solvability check.

Output

Your program should print the minimum number of moves, moved tiles and the solution path.

Sample Input

```
4
1  2  3  4
5  6  0  0
9 10  7  8
13 14 11 12
```

Sample Output

```
Minimum number of moves = 4
Moved tiles: 8 7 12 11

Step 0:
1  2  3  4
5  6  0  0
9 10  7  8
```

```
13 14 11 12

Step 1:
1  2  3  4
5  6  0  8
9 10  7  0
13 14 11 12

Step 2:
1  2  3  4
5  6  7  8
9 10  0  0
13 14 11 12

Step 3:
1  2  3  4
5  6  7  8
9 10  0 12
13 14 11  0

Step 4:
1  2  3  4
5  6  7  8
9 10 11 12
13 14  0  0
```

The printed path may differ if multiple optimal paths exist, but the minimum number of moves should be correct.